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NMAM INSTITUTE OF TECHNOLOGY, NITTE
(An Autonomous Institution affiliated to VTU, Belagavi)
Sixth Semester B.E. (Credit System) Degree Examinations
Supplementary Examinations - December 2025

20MA8X02 - LINEAR ALGEBRA

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I** & **Unit – II** each and **One full** question from **Unit – III**.*

Unit – I		Marks	BT*	CO*	PO*
1.	a) If $V_1, V_2, \dots, V_P$ are in vector space V , then prove that $\text{span}\{V_1, V_2, \dots, V_P\}$ is a subspace of V .	6	L*3	2	1
	b) Prove the Rank Nullity theorem for linear transformation.	7	L3	3	1
	c) Let $G: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be defined by $G\{x, y, z\} = (2y+z, x-4y, 3x)$. Find the matrix A representing G relative to the basis of $\mathbb{R}^3$ as $S = \{(1, 1, 1), (1, 1, 0), (1, 0, 0)\}$	7	L3	3	2
2.	a) Verify Rank-Nullity theorem for the linear transformation $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ defined by $T(x, y, z) = (x+2y-z, y+z, x+y-2z)$.	7	L3	3	1
	b) Determine whether the set S is a linearly dependent subset of $M_{2 \times 2}$ matrices: $S = \left\{ \begin{bmatrix} 2 & 0 \\ -3 & 1 \end{bmatrix}, \begin{bmatrix} -4 & -1 \\ 0 & 5 \end{bmatrix}, \begin{bmatrix} -8 & -3 \\ -6 & 17 \end{bmatrix} \right\}$	7	L2	2	1
	c) Show that the linear transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(x_1, x_2) = (5x_1 - x_2, 2x_1 - 3x_2)$ is invertible and hence find the inverse linear transformation T^{-1} .	6	L3	3	1
3.	a) Find bases for the row space and column space of the matrix $A = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 4 & 5 & 6 \\ 3 & 4 & 5 & 6 & 7 \\ 4 & 5 & 6 & 7 & 8 \end{bmatrix}$	7	L2	2	1
	b) Let W be the subspace of $\mathbb{R}^5$ spanned by the vectors $u_1 = (1, -2, 0, 4)$, $u_2 = (3, 1, 1, 0)$, $u_3 = (-1, -5, -1, 8)$, $u_4 = (3, 8, 2, -12)$. Find a basis and dimension of W .	7	L2	2	2
	c) Check whether the vector $u = (3, -1, 3)$ is in the column space of the matrix $A = \begin{bmatrix} 2 & 4 & -2 & 1 \\ -2 & -5 & 7 & 3 \\ 3 & 7 & -8 & 6 \end{bmatrix}$	6	L2	2	1
Unit – II					
4.	a) Using Cayley-Hamilton theorem find the inverse of the matrix $A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 3 & -3 \\ -2 & -4 & -4 \end{bmatrix}$	7	L2	4	2
	b) Apply Gram-Schmidt orthogonalization process to the vectors. $x_1 = (0, 1, 1, 1)$, $x_2 = (1, 0, 1, 1)$, $x_3 = (1, 1, 0, 1)$, $x_4 = (1, 1, 1, 0)$ to obtain orthonormal basis for the subspace U of $\mathbb{R}^4$.	7	L3	5	2
	c) Show that $U = \{u_1, u_2, u_3\}$ is an orthogonal basis in $\mathbb{R}^3$. Then express x as a linear combination of u where $u_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$, $u_2 = \begin{bmatrix} -1 \\ 4 \\ 1 \end{bmatrix}$, $u_3 = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}$ & $x = \begin{bmatrix} 8 \\ -4 \\ -3 \end{bmatrix}$	6	L2	5	2

P.T.O.

5. a) Find QR factorization of the matrix $\begin{bmatrix} 1 & -1 & -1 \\ 1 & 0 & 0 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \end{bmatrix}$ 10 L3 5 2
- b) Find a least square solution of $AX = B$ where
 $A = \begin{bmatrix} 1 & 3 & 5 \\ 1 & 1 & 0 \\ 1 & 1 & 2 \\ 1 & 3 & 3 \end{bmatrix}$ & $B = \begin{bmatrix} 3 \\ 5 \\ 7 \\ -3 \end{bmatrix}$ 10 L3 5 2
6. a) Find QR factorization of $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ 10 L3 5 2
- b) Find a least square solution of the inconsistent system $AX=b$ for
 $A = \begin{bmatrix} 4 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix}$ & $b = \begin{bmatrix} 2 \\ 0 \\ 11 \end{bmatrix}$ 5 L3 5 2
- c) Find the minimal polynomial of the matrix $A = \begin{bmatrix} 3 & 0 & 1 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$ 5 L3 5 2

Unit – III

7. a) Diagonalize the matrix
 $A = \begin{bmatrix} 8 & -8 & -2 \\ 4 & -3 & -2 \\ 3 & -4 & 1 \end{bmatrix}$ 10 L3 4 3
- b) Find a singular value decomposition of $A = \begin{bmatrix} 4 & 11 & 14 \\ 8 & 7 & -2 \end{bmatrix}$ 10 L3 4 3
8. a) Reduce the quadratic form $x_1^2 + 3x_2^2 + 3x_3^2 - 2x_2x_3$ to the canonical form by an orthogonal transformation. 10 L3 4 1
- b) Find a singular value decomposition of $A = \begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 2 & -2 \end{bmatrix}$ 10 L3 4 3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
